

Complications Associated With Venovenous Extracorporeal Membrane Oxygenation—What Can Go Wrong?

OBJECTIVES: Despite increasing use and promising outcomes, venovenous extracorporeal membrane oxygenation (V-V ECMO) introduces the risk of a number of complications across the spectrum of ECMO care. This narrative review describes the variety of short- and long-term complications that can occur during treatment with ECMO and how patient selection and management decisions may influence the risk of these complications.

DATA SOURCES: English language articles were identified in PubMed using phrases related to V-V ECMO, acute respiratory distress syndrome, severe respiratory failure, and complications.

STUDY SELECTION: Original research, review articles, commentaries, and published guidelines from the Extracorporeal Life support Organization were considered.

DATA EXTRACTION: Data from relevant literature were identified, reviewed, and integrated into a concise narrative review.

DATA SYNTHESIS: Selecting patients for V-V ECMO exposes the patient to a number of complications. Adequate knowledge of these risks is needed to weigh them against the anticipated benefit of treatment. Timing of ECMO initiation and transfer to centers capable of providing ECMO affect patient outcomes. Choosing a configuration that insufficiently addresses the patient's physiologic deficit leads to consequences of inadequate physiologic support. Suboptimal mechanical ventilator management during ECMO may lead to worsening lung injury, delayed lung recovery, or ventilator-associated pneumonia. Premature decannulation from ECMO as lungs recover can lead to clinical worsening, and delayed decannulation can prolong exposure to complications unnecessarily. Short-term complications include bleeding, thrombosis, and hemolysis, renal and neurologic injury, concomitant infections, and technical and mechanical problems. Long-term complications reflect the physical, functional, and neurologic sequelae of critical illness. ECMO can introduce ethical and emotional challenges, particularly when bridging strategies fail.

CONCLUSIONS: V-V ECMO is associated with a number of complications. ECMO selection, timing of initiation, and management decisions impact the presence and severity of these potential harms.

KEY WORDS: acute respiratory distress syndrome; critical care; extracorporeal membrane oxygenation

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The use of venovenous extracorporeal membrane oxygenation (V-V ECMO) in patients with severe respiratory failure reduces the intensity of mechanical ventilation, thereby decreasing ventilator-induced lung injury (1–6). In a subset of patients, ECMO may be lifesaving (7–10). However,

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DOI: 10.1097/CCM.0000000000005673



KEY POINTS

- **Objective:** Provide a narrative review of all potential complications that can arise from the use of venovenous extracorporeal membrane oxygenation for severe respiratory failure.
- **Findings:** V-V ECMO is associated with a considerable number of complications, including those rising from erroneous patient selection, timing of initiation, ECMO configuration, cannulation related complications, and short- and long-term severe critical illness sequelae.
- **Meaning:** An adequate understanding of the implications of ECMO initiation and management decisions, and ECMO-related complications across patients, families, and healthcare teams is crucial to both minimize and address the many patient harms associated with ECMO.

complications related to ECMO are common and affect patient outcomes (10–13). The use of V-V ECMO has increased steadily over time, punctuated by more acute increases during the influenza H1N1 and COVID-19 pandemics (4). The increased demand for V-V ECMO during resource constraints throughout the COVID-19 pandemic led to provision of ECMO outside high-volume tertiary centers, increasing the rate of complications, and worsening outcomes (14, 15). Despite clinical practice guidelines from the Extracorporeal Life Support Organization (16) and data from randomized controlled trials (7), patient selection remains

challenging and requires ECMO complications to be weighed against the anticipated benefit. This narrative review highlights the implications of decision-making around selection, configuration, and removal, and the short- and long-term complications that can occur during treatment with ECMO (Fig. 1).

PATIENT SELECTION

Selection criteria for V-V ECMO are influenced by the inclusion and exclusion criteria of randomized controlled trials and outcome data from large cohort studies (7, 17). Overall, medically eligible patients have a reversible cause of respiratory failure and severe hypoxemia and/or hypercapnia refractory to the maximization of the mechanical ventilator within the limits of lung-protective ventilation and adjunctive therapies such as prone positioning (7). Although no absolute contraindications exist, ECMO is often avoided in patients with an exceedingly poor prognosis (18, 19) or lack of a viable exit strategy (16).

Other characteristics important to the patient's outcome include advanced age, immunocompromised status, severity of hypercapnia and peak inspiratory pressure, cardiac arrest prior to ECMO, and mechanical ventilation days prior to ECMO among others (20–22). However, these data are derived from studies without a comparator group, so although the prognosis of patients with acute respiratory distress syndrome receiving ECMO can be estimated, the margin of benefit provided by ECMO is less clear. Obesity is not considered a contraindication as higher body mass index does not appear to worsen outcomes (22–25).

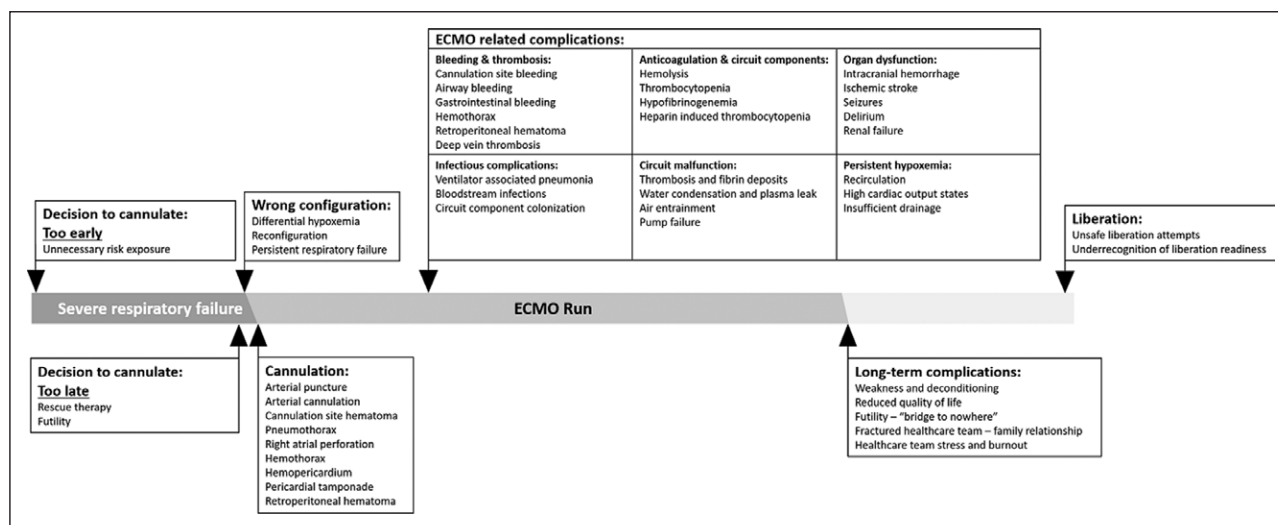


Figure 1. Complications in patients with severe respiratory failure supported with venovenous extracorporeal membrane oxygenation.

However, severe obesity (body mass index $>40\text{ kg/m}^2$) poses important clinical management challenges including prone positioning prior to and during ECMO, obtaining vascular access during cannulation, and achieving the flow rates necessary to meet metabolic demands (24).

Most contraindications are considered relative because no single factor excludes a patient from receiving ECMO, and the importance of these characteristics may vary by the clinical circumstance. For example, evidence from large cohort studies demonstrated worse outcomes among patients for whom ECMO was initiated after 7 days of mechanical ventilation (21, 22, 26). However, a patient who experiences a rapid decline due to a new reversible insult (e.g., ventilator-associated pneumonia) after an initial period of improvement allowing for low-intensity (lung protective) mechanical ventilation may be different from the patient who has received 7 days of high-intensity (potentially injurious) mechanical ventilation. Selecting patients based on intubation days alone may lead to the premature use of ECMO. Therefore, for patients with gas exchange parameters nearing ECMO initiation thresholds, yet tolerating mechanical ventilator settings considered lung protective, we defer ECMO and continue conventional management directed as per clinical guidelines (27).

Ultimately, when selecting patients for ECMO, we consider the margin of benefit ECMO would provide, whether the benefit outweighs the potential ECMO-related complications and whether the patient has clinical characteristics consistent with a reasonable prognosis using ECMO.

TIMING OF ECMO OR TRANSFER TO ECMO CENTER

Data from the Extracorporeal Membrane Oxygenation for Severe Acute Respiratory Distress Syndrome trial suggested that delayed cannulation, as seen in the crossover group, led to higher mortality compared with patients who did not receive ECMO (7). Earlier cannulation along the trajectory of respiratory failure has been associated with improved outcomes in some studies of patients with COVID-19 treated with ECMO (28, 29). However, these differences in outcomes were not consistent when clinicians strictly adhered to lung protective ventilation (30, 31). Likewise, transfer to

centers capable of providing ECMO for patients in whom transport is safe but severe respiratory failure is worsening, facilitates continuous assessment of the patient's clinical trajectory and rapid escalation of treatment when necessary, and has been shown to improve outcomes (17). These data may suggest that: 1) proactive decisions about patient eligibility, especially in the context of injurious levels of mechanical ventilation, may optimize outcomes and 2) attempting to rescue patients after prolonged severe respiratory failure and exposure to high-pressure mechanical ventilation may worsen outcomes.

CONFIGURATION CONSIDERATIONS

The configuration that best meets the patient's physiologic deficit should be selected in order to optimize treatment with ECMO. In our experience, providing the appropriate physiologic platform at the outset has mitigated consequences of inadequate support such as persistent use of injurious ventilator settings or adjunctive therapies, reconfigurations, or circuit modifications.

V-V ECMO provides no direct hemodynamic support; hence, an assessment of hemodynamic conditions and cardiac function is required prior to cannulation, particularly among patients with accompanying hemodynamic instability. In hypoxemic patients with shock receiving high-dose vasopressors in whom echocardiography shows isolated acute cor pulmonale, V-V ECMO allows for rapid correction of hypoxemia, hypercapnia, and reduction of transpulmonary pressures, thereby improving right ventricular afterload and hemodynamics (32, 33). In septic hypoxemic patients whose hemodynamic instability is due to severe left, or biventricular failure, venoarterial (V-A) ECMO provides hemodynamic support and corrects hypoxemia, improving survival (34). However, providers should be aware that additional return cannulas to the venous circulation (V-A-venous ECMO) may be required as a result of differential hypoxemia (35). In contrast, in severely hypoxemic patients with refractory shock in whom biventricular function is normal or mildly reduced, ECMO may be ineffective and may expose patients to complications without any meaningful benefit (34). Erroneously selecting the ECMO configuration may lead to reconfiguration via additional cannula insertion and circuit modifications,

introducing additional complications associated with V-A ECMO, such as differential hypoxemia, left ventricular distension, and limb ischemia—which lay outside the scope of this review (36, 37).

MECHANICAL VENTILATION CONSIDERATIONS

The optimal level of respiratory system unloading in patients supported with V-V ECMO is unknown (5, 38–40). Ventilation strategies during ECMO are heterogeneous (41, 42). Targeting the lowest achievable driving pressure, modest levels of positive end-expiratory pressure, low respiratory rates, and low FiO_2 are recommended (16). However, rapid reduction in lung volumes, apneic ventilation, or total lung collapse may offset the hemodynamic benefits of V-V ECMO in patients with acute cor pulmonale (32, 33). The effect of lung recruitment on lung recovery (via surfactant abnormalities), risk of pneumonia (via atelectasis), inflammation, and survival remains elusive (42–46). In patients with severe hypercapnia prior to cannulation, rapid reduction in ventilatory support may prompt clinicians to use higher levels of sweep gas early after cannulation. Rapid overcorrection of hypercapnia has been shown to increase the risk of neurologic complications in patients supported with ECMO (47, 48). Given the paucity of data informing optimal mechanical ventilator management and risks of rapid and extreme changes to settings, we gradually titrate both mechanical ventilation and ECMO based on the patient's individual physiology and clinical circumstance.

WEANING AND DECANNULATION

Similar to patients on mechanical ventilation, where timely liberation leads to reduced complications and improved outcomes, decannulation at the earliest and safest possible time would be expected to follow a similar path. Evidence on how to wean and liberate patients from V-V ECMO is scarce, and practice patterns vary across centers. Current liberation strategies generally rely on iterative reductions in support and clinician judgment, potentially leading to the underrecognition of readiness for liberation or unsafe liberation attempts. Further, no universal definition of successful decannulation exists (49). Recent data have described standardized approaches to liberation (50, 51), showing promising reductions in ECMO duration (52) and potentially reducing complications and costs.

Deciding when to liberate patients from V-V ECMO is complex, and while more evidence is generated, a standardized approach is needed.

Additional challenges surrounding decannulation occur when bridging strategies fail. ECMO centers should consider pathways for alternative exit strategies in the event of recovery failure (e.g., lung transplant). Otherwise, suboptimal patient management, conflict amongst family and healthcare providers, and strain and stress in healthcare providers and hospital resources may occur.

SHORT-TERM PROBLEMS

Cannulation

Complications related to vascular access are similar to complications observed during the insertion of central lines. Vascular access under ultrasound guidance is the standard of care in most institutions and becomes particularly relevant during the insertion of large bore cannulas (53). Accidental arterial cannulation requires surgical repair and increases the risk of hematoma and retroperitoneal hemorrhage (54). ECMO cannulation requires experienced operators, in particular in obese patients, in whom vessel identification becomes challenging (24). Pneumothorax during cannulation can become a life-threatening situation, as patients are already hypoxemic and receiving high-intensity mechanical ventilation. Echocardiographic guidance can identify guide wire misplacement, pneumothorax, and early right ventricular free-wall perforation (55). Inadvertent arterial cannulation is suspected in the event of acute limb ischemia in patients supported with V-V ECMO (54). Forced blind cannula advancement can result in severe vascular injury, retroperitoneal hemorrhage, hemothorax, or pericardial tamponade (54).

Double-lumen single cannulae are typically inserted in the right internal jugular vein and used to enhance mobilization and conditioning (56). However, disadvantages include blood flow rate restrictions, complex positioning, susceptibility to malpositioning (56), and increased risks of pneumothorax or tamponade at insertion due to right ventricular perforation (57).

Bleeding and Thrombosis

Bleeding and patient and circuit thrombosis are common during V-V ECMO. Thrombosis is more frequent than bleeding, although the occurrence of bleeding has

been associated with a higher mortality than thrombosis (58, 59). Bleeding occurs in 21–66% of patients supported with V-V ECMO (59–61). Although most bleeding complications are minor (59), hemothorax, retroperitoneal hematomas, and gastrointestinal bleeding may cause hemorrhagic shock and death. Bleeding and thrombosis are provoked by exposure of blood to nonbiologic circuit components, causing activation of the coagulation system and degradation of hemostatic factors (62). Differential risk factors for bleeding and thrombosis include age, higher weight, higher pH at initiation, and lower ratio of fraction of Po_2 to Fio_2 (59). Systemic anticoagulation increases the risk of bleeding; however, bleeding may also be observed in patients treated without anticoagulation via thrombocytopenia, hypofibrinogenemia, and shear mediated loss of key platelet surface molecules, selectin, and high-molecular-weight von Willebrand multimers (63). Hemolysis and platelet dysfunction are common and can occur at high- and low-blood flow states, although more data are needed to understand the impact on outcomes (64). Thrombocytopenia is often multifactorial due to sepsis, medications, and the circuit, among others (65, 66). Heparin-induced thrombocytopenia in the treatment of ECMO is rare (3.7%) (66), although carries a high risk of thrombosis to the circuit and patient (2).

Neurologic Complications

Neurologic complications are uncommon in patients receiving ECMO and include seizures, intracranial hemorrhage, ischemic stroke, and brain death. Intracranial hemorrhage is a devastating complication of ECMO affecting 1–6% of cases (59, 67). Risk factors for intracranial hemorrhage include COVID-19 (68, 69) and rapid overcorrection of hypercapnia early after cannulation (47, 48). Survival and functional outcomes after hemorrhagic stroke are generally poor (70–72). Various degrees of ischemic complications, varying from stroke to microvascular thrombosis or microhemorrhages, have been described (73, 74). The mechanisms leading to ischemic neurologic events in patients on V-V ECMO are less clear than for intracranial hemorrhage, and treatment options are limited (59).

Concomitant Infections

Infections are estimated to complicate 30–55% of ECMO runs and impact survival (75, 76). Ventilator-associated

pneumonia is the most frequent nosocomial infection in patients receiving V-V ECMO (76). Blood stream infections in patients on V-V ECMO occur in 13% of ECMO runs and are associated with longer duration of ECMO support (77). ECMO device infection and colonization rates of 10% and 32%, respectively, have been reported (78). The lack of a standardized definition of ECMO-related infections, discerning colonization from infection, and unreliable clinical markers of infection during ECMO makes understanding the relationship between infection and outcomes challenging (75). Further, source control may be difficult due to the impracticalities of exchanging or removing circuit components, leading to longer or recurrent antibiotic treatment courses.

Acute Kidney Injury

Acute kidney injury can occur in up to 60% of patients receiving ECMO (79) and is likely due to concomitant comorbidities and critical illness (79, 80). Acute insults driven by bleeding or sepsis further exacerbate injury mediated by congestive nephropathy and hypoxia in patients with acute cor pulmonale (80, 81). Renal replacement therapy is sometimes unavoidable, and prevention of intradialytic hypotension will help achieve renal recovery (82). Techniques to provide renal replacement therapy while on ECMO may include the addition of an in-line hemofilter to the circuit, a parallel dialysis system (independent vascular access), or integrated systems (dialysis machine connected to the ECMO circuit); each has distinct advantages or disadvantages (80). Severe acute renal failure in patients on V-V ECMO is associated with higher 90-day mortality (83). Prolonged renal replacement therapy (≥ 7 d) is associated with higher ventilator dependence and readmission rates (84).

Circuit Component Malfunctions

Oxygenator or circuit malfunction is the most common ECMO-related complication (59, 67). Membrane aging results in decreased oxygenator efficiency and membrane compartmentalization (85, 86). Oxygenator fibrin deposition/thrombosis, and condensation within the hollow fibers generate membrane dead space and shunt. Degradation of circuit components over time and circuit component exchanges can be problematic, particularly in patients with extremely poor native lung function

and dependent on ECMO for gas exchange. Air entrainment can occur through fractured circuit connectors, sampling ports, or via central access lines in close proximity to the drainage cannula (87). Management generally consists on deairing the premembrane chamber; significant air entrainment will lead to pump air lock or air embolism (87). Other mechanical complications include pump failure, power failure, and inadvertent decannulation. Any interruption to the ECMO circuit to address these complications can lead to rapid gas exchange abnormalities, hemodynamic instability, hemorrhage, and death.

Refractory Hypoxemia

In a subset of patients on V-V ECMO, refractory hypoxemia may develop or persist despite high extracorporeal blood flow. A structured approach is required to identify causes of persistent hypoxemia, including recirculation, high cardiac output states, oxygenator dysfunction, and worsening native lung function (88). Clinically significant recirculation can be identified by reduced color differential in cannulas and high oxygen saturation in premembrane gases (88, 89). High cardiac output states increase the fraction of circulating blood bypassing the ECMO circuit (88, 89). If a mismatch persists between native cardiac output and maximum extracorporeal blood flow, sedation, paralysis, or cooling may reduce oxygen consumption, but may increase delirium, weakness, and infection (88). Beta blockers may reduce intrinsic cardiac output but may be prohibitive due to hypotension and further impairment of tissue oxygen delivery (89). Prone position on ECMO has shown mixed results and introduces management challenges including cannula displacement or dislodgment, endotracheal tube obstruction, and pressure sores (90–92). Higher hemoglobin targets may improve oxygen delivery at the cost of volume overload, infection, and sensitization (88). Additional interventions for refractory hypoxemia that have been used in clinical practice include increasing ventilatory support beyond lung protective ventilation and hemodynamic tolerance, insertion of additional drainage cannulas if blood flow is not already maximized through the single oxygenator, or adding a second oxygenator in series or additional circuit in parallel. Additional mechanical support poses further risk of vascular access complications, coagulopathies, and hemolysis, and evidence

is limited to case series (93, 94). Finally, oxygenation goals on ECMO are controversial, and many ECMO centers target SpO_2 greater than 80–85% while monitoring for evidence of organ dysfunction.

LONG-TERM COMPLICATIONS

Long-term complications and outcomes related to ECMO are vastly underreported and reflect those inherent to critical illness. Nutritional requirements in patients supported with ECMO are undefined; protein dosing in obese and nonobese V-V ECMO patients may be inadequate (95, 96). Early mobilization and rehabilitation in ECMO patients can be limited by hypoxemia, sedation, and hemodynamic instability (97). However, cannulation site is not a contraindication for mobilization (98, 99). Cardiopulmonary dynamics should inform adequate levels of extracorporeal support and titration during physical rehabilitation. When available, patients should be referred to post-ICU follow-up clinics.

Survivors of severe respiratory failure supported with ECMO may experience decrements in health-related quality of life; half return to work (100). The healthcare team should inform patients and families of the potential long-term physical and emotional challenges and promote a shared decision-making model that aligns possible outcomes with patient's expectations and goals of care. ECMO should be introduced to patients and families as a time-limited trial with continuous reassessment of hope toward a meaningful outcome. Without this approach, unrealistic expectations may be generated by patients, family, and the multidisciplinary team. Prolonged nonbeneficial care increases healthcare provider dissatisfaction and burnout (101). Decisions to withdraw ECMO for futility induce significant stress in clinicians, especially in awake patients. Lacking goal-concordant-care results in futility, poor outcomes, and fractured relationships.

CONCLUSIONS

Complications caused, mediated, or compounded by ECMO can occur across the spectrum of ECMO delivery (Fig. 1). Poor patient selection can lead to unnecessary exposure of the broad range of ECMO-related complications or futile care. An inadequate ECMO configuration can lead to insufficient physiologic support requiring injurious adjunctive therapies and avoidable

technical complications. Premature or delayed weaning and liberation from ECMO can lead to recannulation or needless prolonged exposure to ECMO. The use of ECMO can introduce a variety of complications including bleeding, thrombosis, and hemolysis, neurologic and kidney injury, concomitant infections, circuit malfunctions, and others. Long-term complications parallel those of critical illness, including functional, and quality-of-life deficiencies. ECMO can introduce unique ethical challenges and emotional burdens, particularly as bridging strategies fail. An adequate understanding of the implications of ECMO initiation and management decisions, and ECMO-related complications across patients, families, and healthcare teams is crucial to both minimize and address the many patient harms associated with ECMO.

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Dr. Fan reports personal fees from ALung Technologies, Aerogen, Baxter, GE Healthcare, Inspira, and Vasomune outside the submitted work. The authors have disclosed that they do not have any potential conflicts of interest.

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REFERENCES

- Brodie D, Slutsky AS, Combes A: Extracorporeal life support for adults with respiratory failure and related indications: A review. *JAMA* 2019; 322:557–568
- Fan E, Gattinoni L, Combes A, et al: Venovenous extracorporeal membrane oxygenation for acute respiratory failure: A clinical review from an international group of experts. *Intensive Care Med* 2016; 42:712–724
- Slutsky AS, Ranieri VM: Ventilator-induced lung injury. *N Engl J Med* 2013; 369:2126–2136
- Combes A, Schmidt M, Hodgson CL, et al: Extracorporeal life support for adults with acute respiratory distress syndrome. *Intensive Care Med* 2020; 46:2464–2476
- Del Sorbo L, Goffi A, Tomlinson G, et al: International ECMO Network (ECMONet): Effect of driving pressure change during extracorporeal membrane oxygenation in adults with acute respiratory distress syndrome: A randomized crossover physiologic study. *Crit Care Med* 2020; 48:1771–1778
- Rozenowajg S, Guihot A, Franchineau G, et al: Ultra-protective ventilation reduces biotrauma in patients on venovenous extracorporeal membrane oxygenation for severe acute respiratory distress syndrome. *Crit Care Med* 2019; 47:1505–1512
- Combes A, Hajage D, Capellier G, et al; EOLIA Trial Group, REVA, and ECMONet: Extracorporeal membrane oxygenation for severe acute respiratory distress syndrome. *N Engl J Med* 2018; 378:1965–1975
- Combes A, Peek GJ, Hajage D, et al: ECMO for severe ARDS: Systematic review and individual patient data meta-analysis. *Intensive Care Med* 2020; 46:2048–2057
- Goligher EC, Tomlinson G, Hajage D, et al: Extracorporeal membrane oxygenation for severe acute respiratory distress syndrome and posterior probability of mortality benefit in a post hoc Bayesian analysis of a randomized clinical trial. *JAMA* 2018; 320:2251–2259
- Munshi L, Walkey A, Goligher E, et al: Venovenous extracorporeal membrane oxygenation for acute respiratory distress syndrome: A systematic review and meta-analysis. *Lancet Respir Med* 2019; 7:163–172
- Ried M, Sommerauer L, Lubnow M, et al: Thoracic bleeding complications in patients with venovenous extracorporeal membrane oxygenation. *Ann Thorac Surg* 2018; 106:1668–1674
- Sutter R, Tisljar K, Marsch S: Acute neurologic complications during extracorporeal membrane oxygenation: A systematic review. *Crit Care Med* 2018; 46:1506–1513
- Vaquer S, de Haro C, Peruga P, et al: Systematic review and meta-analysis of complications and mortality of veno-venous extracorporeal membrane oxygenation for refractory acute respiratory distress syndrome. *Ann Intensive Care* 2017; 7:51
- Barbaro RP, Odetola FO, Kidwell KM, et al: Association of hospital-level volume of extracorporeal membrane oxygenation cases and mortality. Analysis of the extracorporeal life support organization registry. *Am J Respir Crit Care Med* 2015; 191:894–901
- Gillon SA, Rowland K, Shankar-Hari M, et al: Acceptance and transfer to a regional severe respiratory failure and venovenous extracorporeal membrane oxygenation (ECMO) service: Predictors and outcomes. *Anaesthesia* 2018; 73:177–186
- Tonna JE, Abrams D, Brodie D, et al: Management of adult patients supported with venovenous extracorporeal membrane oxygenation (VV ECMO): Guideline from the Extracorporeal Life Support Organization (ELSO). *ASAIO J* 2021; 67:601–610
- Peek GJ, Mugford M, Tiruvoipati R, et al; CESAR trial collaboration: Efficacy and economic assessment of conventional ventilatory support versus extracorporeal membrane oxygenation for severe adult respiratory failure (CESAR): A multicentre randomised controlled trial. *Lancet* 2009; 374:1351–1363
- Kochanek M, Kochanek J, Böll B, et al: Veno-venous extracorporeal membrane oxygenation (vv-ECMO) for severe respiratory failure in adult cancer patients: A retrospective multicenter analysis. *Intensive Care Med* 2022; 48:332–342
- Schmidt M, Combes A, Shekar K: ECMO for immunosuppressed patients with acute respiratory distress syndrome: Drawing a line in the sand. *Intensive Care Med* 2019; 45:1140–1142
- Pappalardo F, Pieri M, Greco T, et al; Italian ECMONet: Predicting mortality risk in patients undergoing venovenous ECMO for ARDS due to influenza A (H1N1) pneumonia: The ECMONet score. *Intensive Care Med* 2013; 39:275–281

21. Schmidt M, Bailey M, Sheldrake J, et al: Predicting survival after extracorporeal membrane oxygenation for severe acute respiratory failure. The Respiratory Extracorporeal Membrane Oxygenation Survival Prediction (RESP) score. *Am J Respir Crit Care Med* 2014; 189:1374–1382
22. Schmidt M, Zogheib E, Rozé H, et al: The PRESERVE mortality risk score and analysis of long-term outcomes after extracorporeal membrane oxygenation for severe acute respiratory distress syndrome. *Intensive Care Med* 2013; 39:1704–1713
23. Galvagno SM Jr, Pelekhay S, Cornachione CR, et al: Does weight matter? Outcomes in adult patients on venovenous extracorporeal membrane oxygenation when stratified by obesity class. *Anesth Analg* 2020; 131:754–761
24. Javidfar J, Zaaqoq AM, Yamashita MH, et al: Venovenous extracorporeal membrane oxygenation in obese patients. *JTCVS Tech* 2021; 10:335–348
25. Powell EK, Haase DJ, Lankford A, et al: Body mass index does not impact survival in COVID-19 patients requiring venovenous extracorporeal membrane oxygenation. *Perfusion* 2022;2676591221097642
26. Brogan TV, Thiagarajan RR, Rycus PT, et al: Extracorporeal membrane oxygenation in adults with severe respiratory failure: A multi-center database. *Intensive Care Med* 2009; 35:2105–2114
27. Brower RG, Matthay MA, Morris A, et al: Ventilation with lower tidal volumes as compared with traditional tidal volumes for acute lung injury and the acute respiratory distress syndrome. *N Engl J Med* 2000; 342:1301–1308
28. Lebreton G, Schmidt M, Ponnaiah M, et al; Paris ECMO-COVID-19 investigators: Extracorporeal membrane oxygenation network organisation and clinical outcomes during the COVID-19 pandemic in Greater Paris, France: A multicentre cohort study. *Lancet Respir Med* 2021; 9:851–862
29. Urner M, Barnett AG, Bassi GL, et al; COVID-19 Critical Care Consortium Investigators: Venovenous extracorporeal membrane oxygenation in patients with acute covid-19 associated respiratory failure: Comparative effectiveness study. *BMJ* 2022; 377:e068723
30. Diaz RA, Graf J, Zambrano JM, et al: Extracorporeal membrane oxygenation for COVID-19-associated severe acute respiratory distress syndrome in Chile: A nationwide incidence and cohort study. *Am J Respir Crit Care Med* 2021; 204:34–43
31. Hermann M, Laxar D, Krall C, et al: Duration of invasive mechanical ventilation prior to extracorporeal membrane oxygenation is not associated with survival in acute respiratory distress syndrome caused by coronavirus disease 2019. *Ann Intensive Care* 2022; 12:6
32. Bunge JJH, Caliskan K, Gommers D, et al: Right ventricular dysfunction during acute respiratory distress syndrome and veno-venous extracorporeal membrane oxygenation. *J Thorac Dis* 2018; 10:S674–S682
33. Reis Miranda D, van Thiel R, Brodie D, et al: Right ventricular unloading after initiation of venovenous extracorporeal membrane oxygenation. *Am J Respir Crit Care Med* 2015; 191:346–348
34. Ling RR, Ramanathan K, Poon WH, et al: Venoarterial extracorporeal membrane oxygenation as mechanical circulatory support in adult septic shock: A systematic review and meta-analysis with individual participant data meta-regression analysis. *Crit Care* 2021; 25:246
35. Falk L, Sallisalimi M, Lindholm JA, et al: Differential hypoxemia during venoarterial extracorporeal membrane oxygenation. *Perfusion* 2019; 34:22–29
36. Roumy A, Liaudet L, Rusca M, et al: Pulmonary complications associated with veno-arterial extra-corporeal membrane oxygenation: A comprehensive review. *Crit Care* 2020; 24:212
37. Son AY, Khanh LN, Joung HS, et al: Limb ischemia and bleeding in patients requiring venoarterial extracorporeal membrane oxygenation. *J Vasc Surg* 2021; 73:593–600
38. Abrams D, Schmidt M, Pham T, et al: Mechanical ventilation for acute respiratory distress syndrome during extracorporeal life support. Research and practice. *Am J Respir Crit Care Med* 2020; 201:514–525
39. Fan E: “There is nothing new except what has been forgotten”: The story of mechanical ventilation during extracorporeal support. *Am J Respir Crit Care Med* 2019; 199:550–553
40. Quintel M, Busana M, Gattinoni L: Breathing and ventilation during extracorporeal membrane oxygenation: How to find the balance between rest and load. *Am J Respir Crit Care Med* 2019; 200:954–956
41. Marhong JD, Telesnicki T, Munshi L, et al: Mechanical ventilation during extracorporeal membrane oxygenation. An international survey. *Ann Am Thorac Soc* 2014; 11:956–961
42. Schmidt M, Pham T, Arcadipane A, et al: Mechanical ventilation management during extracorporeal membrane oxygenation for acute respiratory distress syndrome. An international multicenter prospective cohort. *Am J Respir Crit Care Med* 2019; 200:1002–1012
43. Araos J, Alegria L, Garcia A, et al: Effect of positive end-expiratory pressure on lung injury and haemodynamics during experimental acute respiratory distress syndrome treated with extracorporeal membrane oxygenation and near-apnoeic ventilation. *Br J Anaesth* 2021; 127:807–814
44. van Kaam AH, Lachmann RA, Herting E, et al: Reducing atelectasis attenuates bacterial growth and translocation in experimental pneumonia. *Am J Respir Crit Care Med* 2004; 169:1046–1053
45. Albert RK: The role of ventilation-induced surfactant dysfunction and atelectasis in causing acute respiratory distress syndrome. *Am J Respir Crit Care Med* 2012; 185:702–708
46. Schmidt M, Stewart C, Bailey M, et al: Mechanical ventilation management during extracorporeal membrane oxygenation for acute respiratory distress syndrome: A retrospective international multicenter study. *Crit Care Med* 2015; 43:654–664
47. Cavayas YA, Munshi L, Del Sorbo L, et al: The early change in PaCO₂ after extracorporeal membrane oxygenation initiation is associated with neurological complications. *Am J Respir Crit Care Med* 2020; 201:1525–1535
48. Luyt CE, Bréchet N, Demondion P, et al: Brain injury during venovenous extracorporeal membrane oxygenation. *Intensive Care Med* 2016; 42:897–907
49. Al-Fares AA, Ferguson ND, Ma J, et al: Achieving safe liberation during weaning from VV-ECMO in patients with severe ARDS: The role of tidal volume and inspiratory effort. *Chest* 2021; 160:1704–1713
50. Gannon WD, Stokes JW, Bloom S, et al: Safety and feasibility of a protocolized daily assessment of readiness for liberation from venovenous extracorporeal membrane oxygenation. *Chest* 2021; 160:1693–1703

51. Teijeiro-Paradis R, Tiwari P, Spriel A, et al: Standardized liberation trials in patients with COVID-19 ARDS treated with venovenous extracorporeal membrane oxygenation: When ready, let them breathe! *Intensive Care Med* 2021; 47:1494–1496
52. Pratt EH, Mausert S, Wilson MD, et al: A daily, respiratory therapist assessment of readiness to liberate from venovenous extracorporeal membrane oxygenation in patients with acute respiratory distress syndrome. *Crit Care Explor* 2021; 3:e0584
53. Leibowitz A, Oren-Grinberg A, Matyal R: Ultrasound guidance for central venous access: Current evidence and clinical recommendations. *J Intensive Care Med* 2020; 35:303–321
54. Rupprecht L, Lunz D, Philipp A, et al: Pitfalls in percutaneous ECMO cannulation. *Heart Lung Vessel* 2015; 7:320–326
55. Castro DM, Morris I, Teijeiro-Paradis R, et al: Monitoring during extracorporeal membrane oxygenation. *Curr Opin Crit Care* 2022; 28:348–359
56. Jayaraman AL, Cormican D, Shah P, et al: Cannulation strategies in adult veno-arterial and veno-venous extracorporeal membrane oxygenation: Techniques, limitations, and special considerations. *Ann Card Anaesth* 2017; 20:S11–S18
57. Rubino A, Vuylsteke A, Jenkins DP, et al: Direct complications of the Avalon bicaval dual-lumen cannula in respiratory extracorporeal membrane oxygenation (ECMO): Single-center experience. *Int J Artif Organs* 2014; 37:741–747
58. Aubron C, DePuydt J, Belon F, et al: Predictive factors of bleeding events in adults undergoing extracorporeal membrane oxygenation. *Ann Intensive Care* 2016; 6:97
59. Nunez JI, Gosling AF, O'Gara B, et al: Bleeding and thrombotic events in adults supported with venovenous extracorporeal membrane oxygenation: An ELSO registry analysis. *Intensive Care Med* 2022; 48:213–224
60. Northam KA, Murray BP, Fischer WA, et al: Major bleeding and thrombosis events in COVID-19 versus influenza patients requiring extracorporeal membrane oxygenation. *ASAIO J* 2022; 68:779–784
61. Willers A, Swol J, Buscher H, et al: Longitudinal trends in bleeding complications on extracorporeal life support over the past two decades-extracorporeal life support organization registry analysis. *Crit Care Med* 2022; 50:e569–e580
62. Doyle AJ, Hunt BJ: Current understanding of how extracorporeal membrane oxygenators activate haemostasis and other blood components. *Front Med (Lausanne)* 2018; 5:352
63. Olson SR, Murphree CR, Zonies D, et al: Thrombosis and bleeding in extracorporeal membrane oxygenation (ECMO) without anticoagulation: A systematic review. *ASAIO J* 2021; 67:290–296
64. Dufour N, Radjou A, Thuong M: Hemolysis and plasma free hemoglobin during extracorporeal membrane oxygenation support: From clinical implications to laboratory details. *ASAIO J* 2020; 66:239–246
65. Abrams D, Baldwin MR, Champion M, et al: Thrombocytopenia and extracorporeal membrane oxygenation in adults with acute respiratory failure: A cohort study. *Intensive Care Med* 2016; 42:844–852
66. Jiritano F, Serraino GF, Ten Cate H, et al: Platelets and extra-corporeal membrane oxygenation in adult patients: A systematic review and meta-analysis. *Intensive Care Med* 2020; 46:1154–1169
67. Barbaro RP, MacLaren G, Boonstra PS, et al: Extracorporeal Life Support Organization: Extracorporeal membrane oxygenation support in COVID-19: An international cohort study of the Extracorporeal Life Support Organization registry. *Lancet* 2020; 396:1071–1078
68. Doyle AJ, Hunt BJ, Sanderson B, et al: A comparison of thrombosis and hemorrhage rates in patients with severe respiratory failure due to coronavirus disease 2019 and influenza requiring extracorporeal membrane oxygenation. *Crit Care Med* 2021; 49:e663–e672
69. Seeliger B, Doebler M, Hofmaenner DA, et al: Intracranial hemorrhages on extracorporeal membrane oxygenation: Differences between COVID-19 and other viral acute respiratory distress syndrome. *Crit Care Med* 2022; 50:e526–e538
70. Fletcher-Sandersjö A, Thelin EP, Bartek J Jr, et al: Management of intracranial hemorrhage in adult patients on extracorporeal membrane oxygenation (ECMO): An observational cohort study. *PLoS One* 2017; 12:e0190365
71. Hunsicker O, Beck L, Krannich A, et al: Timing, outcome, and risk factors of intracranial hemorrhage in acute respiratory distress syndrome patients during venovenous extracorporeal membrane oxygenation. *Crit Care Med* 2021; 49:e120–e129
72. Prinz V, Manekeller L, Menk M, et al: Clinical management and outcome of adult patients with extracorporeal life support device-associated intracerebral hemorrhage—a neurocritical perspective and grading. *Neurosurg Rev* 2021; 44:2879–2888
73. Lorusso R, Belliato M, Mazzeffi M, et al: Neurological complications during veno-venous extracorporeal membrane oxygenation: does the configuration matter? A retrospective analysis of the ELSO database. *Crit Care* 2021; 25:107
74. Marinoni M, Migliaccio ML, Trapani S, et al: Cerebral micro-emboli detected by transcranial Doppler in patients treated with extracorporeal membrane oxygenation. *Acta Anaesthesiol Scand* 2016; 60:934–944
75. Abrams D, Grasselli G, Schmidt M, et al: ECLS-associated infections in adults: What we know and what we don't yet know. *Intensive Care Med* 2020; 46:182–191
76. Grasselli G, Scaravilli V, Di Bella S, et al: Nosocomial infections during extracorporeal membrane oxygenation: Incidence, etiology, and impact on patients' outcome. *Crit Care Med* 2017; 45:1726–1733
77. Menaker J, Galvagno S, Rabinowitz R, et al: Epidemiology of blood stream infection in adult extracorporeal membrane oxygenation patients: A cohort study. *Heart Lung* 2019; 48:236–239
78. Thomas G, Hraiech S, Cassir N, et al: Venovenous extracorporeal membrane oxygenation devices-related colonisations and infections. *Ann Intensive Care* 2017; 7:111
79. Pilarczyk K, Huenges K, Bewig B, et al: Acute kidney injury in patients with severe ARDS requiring extracorporeal membrane oxygenation: Incidence, prognostic impact and risk factors. *J Clin Med* 2022; 11:1079
80. Ostermann M, Lumlertgul N: Acute kidney injury in ECMO patients. *Crit Care* 2021; 25:313
81. Husain-Syed F, Ricci Z, Brodie D, et al: Extracorporeal organ support (ECOS) in critical illness and acute kidney injury: From

- native to artificial organ crosstalk. *Intensive Care Med* 2018; 44:1447–1459
82. Douvris A, Zeid K, Hiremath S, et al: Mechanisms for hemodynamic instability related to renal replacement therapy: A narrative review. *Intensive Care Med* 2019; 45:1333–1346
 83. Delmas C, Zapetskaia T, Conil JM, et al: 3-month prognostic impact of severe acute renal failure under veno-venous ECMO support: Importance of time of onset. *J Crit Care* 2018; 44:63–71
 84. Kuo G, Chen SW, Fan PC, et al: Analysis of survival after initiation of continuous renal replacement therapy in patients with extracorporeal membrane oxygenation. *BMC Nephrol* 2019; 20:318
 85. Epis F, Belliato M: Oxygenator performance and artificial-native lung interaction. *J Thorac Dis* 2018; 10:S596–S605
 86. Zakhary B, Sheldrake J, Pellegrino V: Extracorporeal membrane oxygenation and V/Q ratios: An ex vivo analysis of CO₂ clearance within the Maquet Quadrox-iD oxygenator. *Perfusion* 2020; 35:29–33
 87. Chan KM, Wan WTP, Ling L, et al: Management of circuit air in extracorporeal membrane oxygenation: A single center experience. *ASAIO J* 2022; 68:e39–e43
 88. Montisci A, Maj G, Zangrillo A, et al: Management of refractory hypoxemia during venovenous extracorporeal membrane oxygenation for ARDS. *ASAIO J* 2015; 61:227–236
 89. Bartlett RH: Physiology of gas exchange during ECMO for respiratory failure. *J Intensive Care Med* 2017; 32:243–248
 90. Giani M, Rezoagli E, Guervilly C, et al; EuroPronECMO Investigators: Prone positioning during venovenous extracorporeal membrane oxygenation for acute respiratory distress syndrome: A pooled individual patient data analysis. *Crit Care* 2022; 26:8
 91. Papazian L, Schmidt M, Hajage D, et al: Effect of prone positioning on survival in adult patients receiving venovenous extracorporeal membrane oxygenation for acute respiratory distress syndrome: A systematic review and meta-analysis. *Intensive Care Med* 2022; 48:270–280
 92. Poon WH, Ramanathan K, Ling RR, et al: Prone positioning during venovenous extracorporeal membrane oxygenation for acute respiratory distress syndrome: A systematic review and meta-analysis. *Crit Care* 2021; 25:292
 93. Patel YJ, Stokes JW, Gannon WD, et al: Extracorporeal membrane oxygenation circuits in parallel for refractory hypoxemia in COVID-19: A case series. *ASAIO J* 2022; 68:1002–1009
 94. Shah A, Dave S, Goerlich CE, et al: Hybrid and parallel extracorporeal membrane oxygenation circuits. *JTCVS Tech* 2021; 8:77–85
 95. MacGowan L, Smith E, Elliott-Hammond C, et al: Adequacy of nutrition support during extracorporeal membrane oxygenation. *Clin Nutr* 2019; 38:324–331
 96. Pelekhaty SL, Galvagno SM Jr, Lantry JH, et al: Are current protein recommendations for the critically ill adequate for patients on VV ECMO: Experience from a high-volume center. *JPEN J Parenter Enteral Nutr* 2020; 44:220–226
 97. Hodgson CL, Hayes K, Linnane M, et al: Early mobilisation during extracorporeal membrane oxygenation was safe and feasible: a pilot randomised controlled trial. *Intensive Care Med* 2020; 46:1057–1059
 98. Abrams D, Madahar P, Eckhardt CM, et al; MORE-PT Investigators: Early mobilization during extracorporeal membrane oxygenation for cardiopulmonary failure in adults: Factors associated with intensity of treatment. *Ann Am Thorac Soc* 2022; 19:90–98
 99. Munshi L, Kobayashi T, DeBacker J, et al: Intensive care physiotherapy during extracorporeal membrane oxygenation for acute respiratory distress syndrome. *Ann Am Thorac Soc* 2017; 14:246–253
 100. Wilcox ME, Jaramillo-Rocha V, Hodgson C, et al: Long-term quality of life after extracorporeal membrane oxygenation in ARDS survivors: Systematic review and meta-analysis. *J Intensive Care Med* 2020; 35:233–243
 101. Lambden JP, Chamberlin P, Kozlov E, et al: Association of perceived futile or potentially inappropriate care with burnout and thoughts of quitting among health-care providers. *Am J Hosp Palliat Care* 2019; 36:200–206